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Course: BE-CSE (AI&ML)

Subject: Database Management System

Experiment 6: Working of Cursor in PL/SQL

1. Aim of the Session

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyse how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

2. Software Requirements:

- Database Management System:
  - Oracle Database Express Edition (Oracle XE)
  - PostgreSQL Database

- Database Administration Tool / Client Tool:
  - Oracle SQL Developer (for Oracle XE)
  - pgAdmin (for PostgreSQL)

3. Objective of the Session

To implement and analyse the use of implicit cursors, explicit cursors, and cursor attributes for processing multiple rows from a database table and applying business logic effectively.

4. Practical / Experiment Steps

The work was carried out through the following activities:

1. **Schema Creation**: Created the Employees table with attributes such as Employee ID, Name, Role, and Salary to represent a structured organizational database.
2. **Data Initialization:** Inserted multiple employee records with different roles and salary values to prepare a dataset for cursor operations.

3. **Implicit Cursor Implementation:** Executed an anonymous PL/SQL block to perform an UPDATE operation on employee salary and used implicit cursor attributes like SQL%ROWCOUNT to verify execution.
4. **Explicit Cursor Declaration:** Defined a named cursor (c_emp) to retrieve records of employees based on a specific condition (e.g., role = 'Engineer') for controlled row-wise processing.
5. **Cursor Attribute Utilization:** Applied cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN to manage program flow and monitor cursor behavior.
6. **Logical Processing:** Implemented business logic to process employee records individually, such as updating salaries or displaying formatted output.

5. Procedure of the Practical

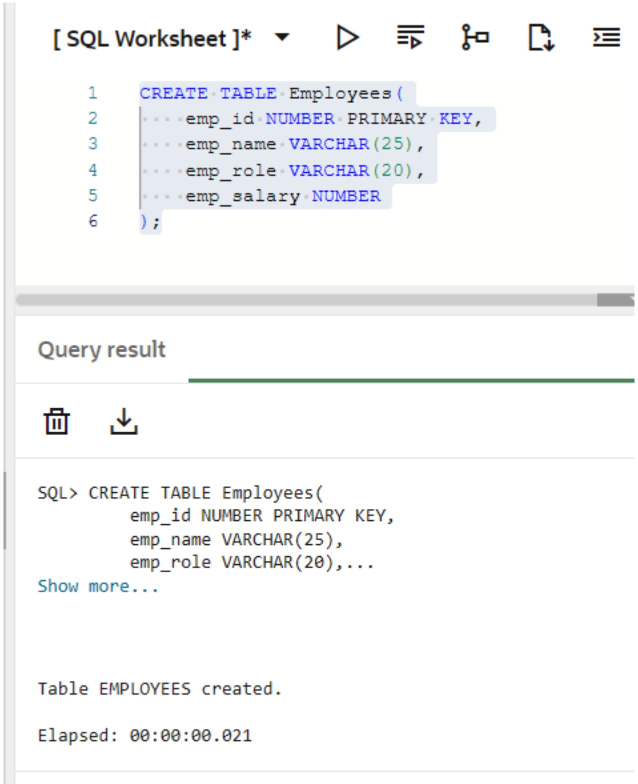
Execution was performed in the following order:

- Opened the Oracle database environment (SQL*Plus / SQL Developer) and enabled SERVEROUTPUT to display results.
- Executed CREATE TABLE and INSERT statements to establish the Employees dataset.
- Ran a PL/SQL block using an implicit cursor to update a selected employee’s salary and verified the result using SQL%ROWCOUNT.
- Declared an explicit cursor (c_emp) to select employees based on a specific role condition.
- Opened the cursor and confirmed its status using the %ISOPEN attribute.
- Used a LOOP structure to fetch records sequentially into variables (e.g., v_name, v_salary).
- Applied %NOTFOUND as the exit condition to terminate the loop after all records were processed.
- Tracked the number of processed rows using %ROWCOUNT during iteration.
- Closed the cursor explicitly to release system resources and verified closure using NOT c_emp%ISOPEN.

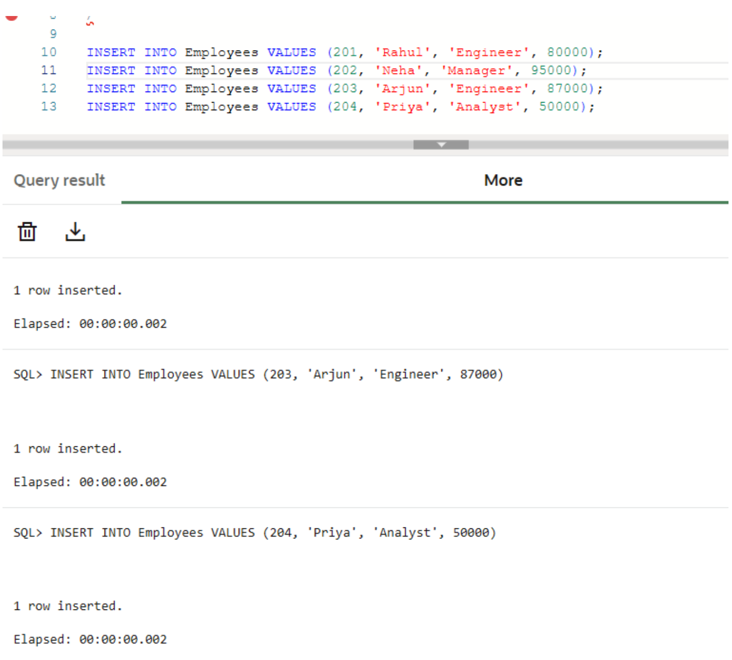
6. I/O Analysis (Input / Output Analysis)

Input Queries

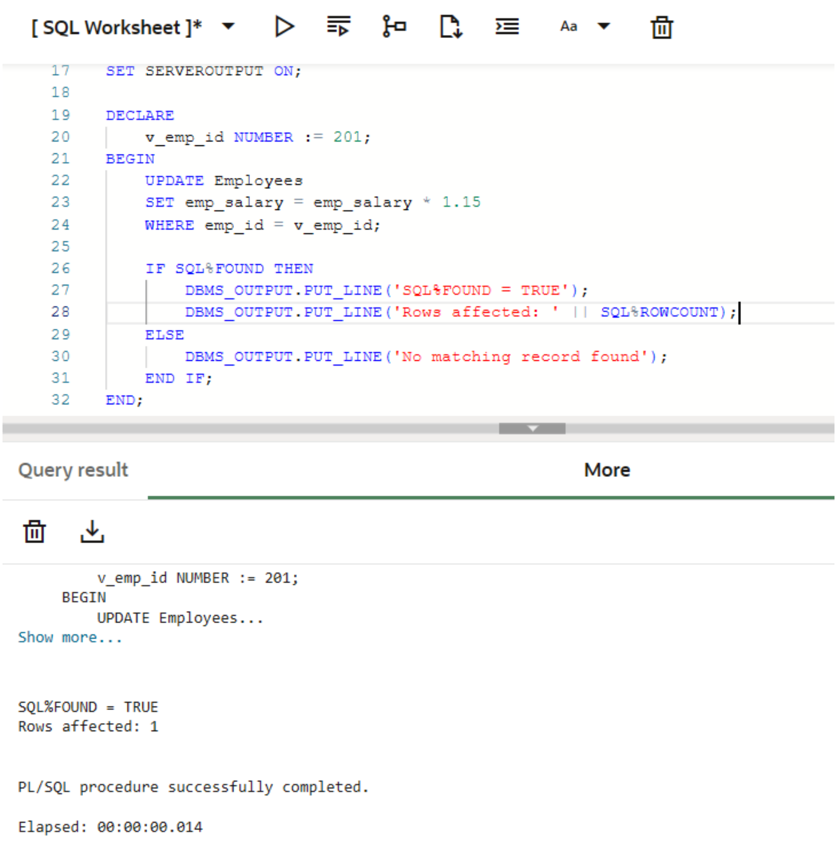
SQL
```sql
CREATE TABLE Employees(
    emp_id NUMBER PRIMARY KEY,
    emp_name VARCHAR(25),
    emp_role VARCHAR(20),
    emp_salary NUMBER
);
```




```sql
INSERT INTO Employees VALUES (201, 'Rahul', 'Engineer', 80000);
INSERT INTO Employees VALUES (202, 'Neha', 'Manager', 95000);
INSERT INTO Employees VALUES (203, 'Arjun', 'Engineer', 87000);
INSERT INTO Employees VALUES (204, 'Priya', 'Analyst', 50000);
```

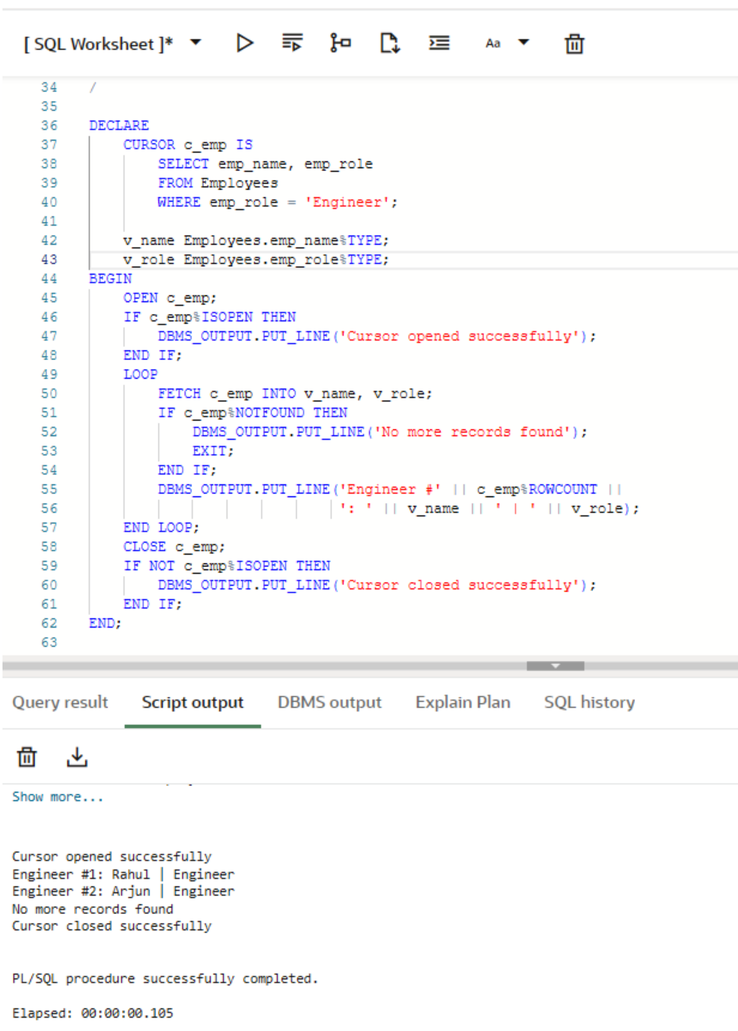


```
SET SERVEROUTPUT ON;

DECLARE
    v_emp_id NUMBER := 201;
BEGIN
    UPDATE Employees
    SET emp_salary = emp_salary * 1.15
    WHERE emp_id = v_emp_id;

    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('SQL%FOUND = TRUE');
        DBMS_OUTPUT.PUT_LINE('Rows affected: ' || SQL%ROWCOUNT);
    ELSE
        DBMS_OUTPUT.PUT_LINE('No matching record found');
    END IF;
END;
```




```
DECLARE
    CURSOR c_emp IS
        SELECT emp_name, emp_role
        FROM Employees
        WHERE emp_role = 'Engineer';

    v_name Employees.emp_name%TYPE;
    v_role Employees.emp_role%TYPE;
```

```
BEGIN
    OPEN c_emp;
    IF c_emp%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor opened successfully');
    END IF;
    LOOP
        FETCH c_emp INTO v_name, v_role;
        IF c_emp%NOTFOUND THEN
            DBMS_OUTPUT.PUT_LINE('No more records found');
            EXIT;
        END IF;
        DBMS_OUTPUT.PUT_LINE('Engineer #' || c_emp%ROWCOUNT ||
                            ': ' || v_name || ' | ' || v_role);
    END LOOP;
    CLOSE c_emp;
    IF NOT c_emp%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor closed successfully');
    END IF;
END;
```




```
/

DECLARE
    CURSOR c_emp IS
        SELECT emp_name, emp_role
        FROM Employees
        WHERE emp_role = 'Engineer';

    v_name Employees.emp_name%TYPE;
    v_role Employees.emp_role%TYPE;
BEGIN
    OPEN c_emp;
    IF c_emp%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor opened successfully');
    END IF;
    LOOP
        FETCH c_emp INTO v_name, v_role;
        IF c_emp%NOTFOUND THEN
            DBMS_OUTPUT.PUT_LINE('No more records found');
            EXIT;
        END IF;
        DBMS_OUTPUT.PUT_LINE('Engineer #' || c_emp%ROWCOUNT ||
                            ': ' || v_name || ' | ' || v_role);
    END LOOP;
    CLOSE c_emp;
    IF NOT c_emp%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor closed successfully');
    END IF;
END;
```

```
Show more...

Cursor opened successfully
Engineer #1: Rahul | Engineer
Engineer #2: Arjun | Engineer
No more records found
Cursor closed successfully

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.105
```

7. Learning Outcome

- Learned how to use implicit and explicit cursors in PL/SQL.
- Understood cursor attributes like `SQL%ROWCOUNT`, `%FOUND`, `%NOTFOUND`, and `%ISOPEN`.
- Gained ability to process records row-by-row using cursors.
- Applied simple logic on database records (update and display).